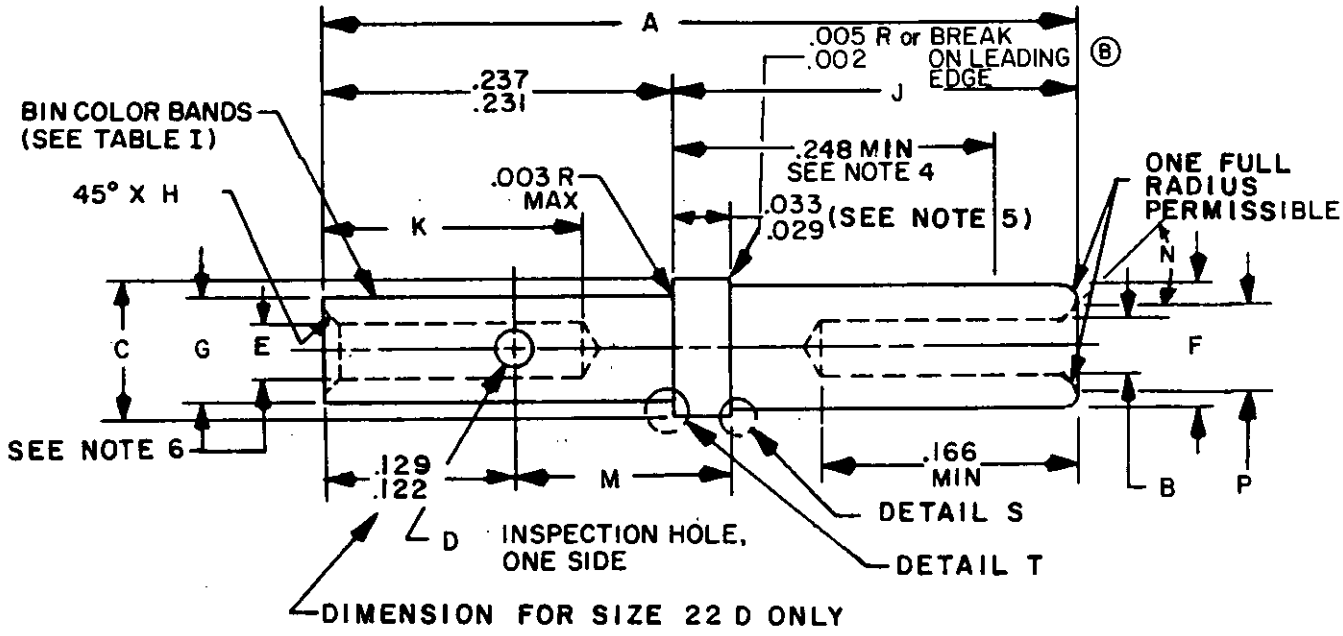
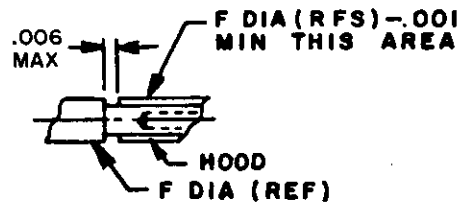
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, (THERMOCOUPLE FOR MIL-C-38999 SERIES II, MIL-C-83733, AND MIL-C-24308 CONNECTORS)

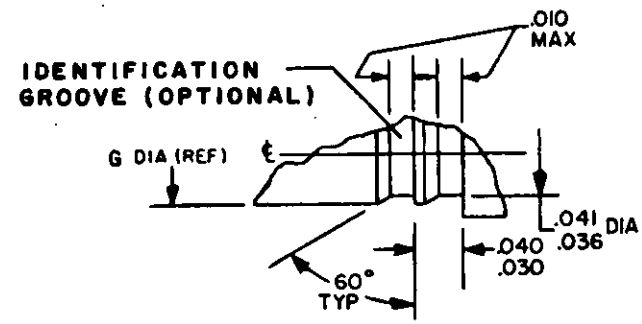
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.





DETAIL S (ALL SIZES)



DETAIL T SIZE 22 D ONLY

FIGURE 1. Socket contacts.

Ⓑ denotes change

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INCHES	MM	INCHES	MM	INCHES	MM
.002	0.05	.044	1.12	.088	2.24
.003	0.08	.046	1.17	.091	2.31
.005	0.13	.047	1.19	.094	2.39
.010	0.25	.048	1.22	.101	2.57
.018	0.46	.053	1.35	.103	2.62
.022	0.56	.060	1.52	.113	2.87
.026	0.66	.062	1.57	.127	3.23
.029	0.74	.064	1.63	.130	3.30
.031	0.79	.066	1.68	.141	3.58
.032	0.81	.068	1.73	.209	5.31
.0335	0.85	.070	1.78	.231	5.87
.0355	0.90	.072	1.83	.237	6.02
.036	0.91	.078	1.98	.279	7.09
.0415	1.05	.082	2.08	.289	7.34
.042	1.07	.084	2.16	.518	13.16

BIN code	A REF	B DIA	C DIA	D DIA	E DIA	F DIA MAX	G DIA	H	J	K MIN	M	N°	P DIA MIN
494 through 497	.518	.031	.062	.022	.0355	.062	.048	.005	.289	.141	---	50°	.047
		MIN	.060	.018	.0335		.046	.003				44°	
498 through 501		.0415	.094	.032	.048	.078	.070	.010		.209	.078	47°	.053
		MIN	.091	.026	.046		.068	.005	.279		.072	28°	
502 through 505		.064	.130	.042	.068	.113	.103	.010			.088		.084
		MIN	.127	.036	.066		.101	.005			.082		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Point at which a square ended pin of the same basic diameter as the mating contact first engages the socket contact spring.
5. Indicated dimension does not apply for size 22D.
6. For size 22D only, diameters E and G to be concentric within .003 (TIR) regardless of feature size (RFS); for all other contact sizes, diameters E and G to be concentric within .001 (TIR) at maximum material condition (MMC).
7. Hoods shall not exceed contact body diameter regardless of feature size (RFS) in attachment area.

FIGURE 1. Socket contacts - Continued.

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TABLE I. Design characteristics.

BIN code	Color bands			Mating end size	Wire barrel size	Type	Class
	1st	2nd	3rd				
494	Yellow	White	Yellow	22	22D	C (See table III)	B
495	Yellow	White	Green	22	22D		
496	Yellow	White	Blue	22	22D		
497	Yellow	White	Violet	22	22D		
498	Yellow	White	Gray	20	20		
499	Yellow	White	White	20	20		
500	Green	Black	Black	20	20		
501	Green	Black	Brown	20	20		
502	Green	Black	Red	16	16		
503	Green	Black	Orange	16	16		
504	Green	Black	Yellow	16	16		
505	Green	Black	Green	16	16		

TABLE II. Tools.

BIN code	Basic crimping tool	Positioner	Installing tool	Removal tool
494	M22520/2-01	M22520/2-06	M81969/14-01	M81969/14-01
495	M22520/7-01	M22520/7-06	M81969/8-01	M81969/8-02
496				
497				
498	M22520/1-01	M22520/1-04	M81969/8-05	M81969/8-06
499	M22520/2-01	M22520/2-10	M81969/14-02	M81969/14-02
500	M22529/7-01	M22520/7-08		
501				
502	M22520/1-01	M22520/1-04	M81969/8-07	M81969/8-08
503	M22520/7-01	M22520/7-04	M81969/14-03	M81969/14-03
504				
505				

TABLE III. Materials, plating, and tensile strength. Ⓑ

BIN code	Material	Plating	Tensile strength 1/			
			Wire size	Axial load	Wire size	Axial load
494	Constantan	None	28	3 lbf	22	8 lbf
495	Alumel	None		4 lbf		9.25 lbf
496	Chromel	None		6 lbf		11.25 lbf
497	Iron	Cadmium plate 2/		3 lbf		8 lbf
498	Constantan	None	24	10 lbf	20	55 lbf
499	Alumel	None		15 lbf		60 lbf
500	Chromel	None		18 lbf		65 lbf
501	Iron	Cadmium plate 2/		10 lbf		55 lbf
502	Constantan	None	20	45 lbf	16	135 lbf
503	Alumel	None		55 lbf		145 lbf
504	Chromel	None		60 lbf		150 lbf
505	Iron	Cadmium plate 2/		45 lbf		135 lbf

1/ Applies to type 1 wire per MIL-W-5846.
2/ Chromate clear coat.

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REQUIREMENTS:

Qualification: Contacts shall comply with the reliability assurance provisions at MIL-STD-790 as specified in MIL-C-38999.

Dimensions, design characteristics, and configuration: See figure 1 and table 1.

Mating contacts: MIL-C-39029/87.

Tools: See table II.

Materials and plating: See table III.

Tensile strength: See table III.

Vibration: Test condition VI, letter J of method 2005 of MIL-STD-1344, except duration shall be 8 hours in the longitudinal direction and 8 hours in the perpendicular direction (16 hours total).

Shock (specified pulse): Test condition D of method 2004 of MIL-STD-1344, except the pulse shall be an approximate half sine wave of 300 g ±15 percent magnitude with a duration of 3 milliseconds ±1.

Resistance to test probe damage: See table IV.

TABLE IV. Probe damage insertion depths.

Contact size	Insertion depth +.020 - .000 inches
22D, 20	.083 - .125 1/
16	.135
12	1.66

1/ Indicated dimensions represent 1/2 and 3/4 depths.

QPL evaluating activity: Defense Electronics Supply Center (DESC-E), Dayton, Ohio, 45444.

International interest: NEPR 57.

Part numbers: See table V.

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TABLE V. Part number and BIN code.

BIN code	Part number	Superseded
494	M39029/89-494	M39029/892222C1
495	M39029/89-495	M39029/892222C2
496	M39029/89-496	M39029/892222C3
497	M39029/89-497	M39029/892222C4
498 1/	M39029/89-498	M39029/892020C1
499 1/	M39029/89-499	M39029/892020C2
500 1/	M39029/89-500	M39029/892020C3
501 1/	M39029/89-501	M39029/892020C4
502 1/	M39029/89-502	M39029/891616C1
503 1/	M39029/89-503	M39029/891616C2
504 1/	M39029/89-504	M39029/891616C3
505 1/	M39029/89-505	M39029/891616C4

1/ Not for use with MIL-C-83733 or MIL-C-24308 connectors (for MIL-C-83733 contact sizes 20 and 16, see MIL-C-39029/9 and MIL-C-39029/10).

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR
Navy - EC
Air Force - 11, 99
DLA - ES

User activities:
Navy - OS, SH

Agent:
DLA - ES

Preparing activity:
Air Force - 85

(Project 5935-3474-3)

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